# Angle Pairs: Supplementary, Complementary, Vertical, Adjacent

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 6 — Relationships in Triangles |
| Lesson | Thread B · Day 1 — Angle Pairs: Supplementary, Complementary, Vertical, Adjacent | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.8.GR.1.4 |

## 2. Standards

**MA.8.GR.1.4** Solve mathematical problems involving the relationships between supplementary, complementary, vertical or adjacent angles.

**Must.** Name adjacent, vertical, complementary and supplementary pairs (a linear pair is supplementary); find a missing angle from each; write and solve the equation when the angles are expressions, and report the angle the question asks for.

**Must not.** Parallel lines and transversals (Geometry); angle sums of polygons (MA.8.GR.1.6); reflex angles.

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.2.1 | Demonstrate understanding by representing problems in multiple ways. | Notes II–III: one pair as a picture, a name and an equation |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 4 and 7 split; the re-vote |
| MA.K12.MTR.5.1 | Use patterns and structure to help understand and connect mathematical concepts. | Notes III and boards 5–6: the same equation shape behind every pair |

## 4. Learning Target and Essential Question

**Learning target.** I can name an angle pair and use it to find a missing angle, with an equation when the angles are expressions.

**Essential question.** When two angles are neighbours or opposites, what does one tell you about the other?

**Building on.** Grade 4 (MA.4.GR.1.3) — angle measures add, a straight angle is 180°; Unit 1 — two-step equations (MA.7.AR.2.2). An x on both sides is new: Example 2 takes it off one side as its own line.

**Working toward.** T-B2 tomorrow — the angles of a triangle; Geometry — angle pairs made by parallel lines and a transversal.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 10 | Teach from the front; say the OFF line out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 3 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 12 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; finish tonight. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (12 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| Can two adjacent angles be vertical angles? Why or why not? | DOK 2 |
| Two angles are supplementary and equal. What are they, and what do they make? | DOK 2 |
| Board 6's x is 12 and its angles are 52°. Which does the question ask for, and how do you know? | DOK 2 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | What is the complement of 34°? | ***56°*** | 146° — used 180 [Florida B1G-M, GR.1.4: complementary and supplementary swapped] |
| 2 | What is the supplement of 112°? | ***68°*** | −22 or 'none' — used 90 [Florida B1G-M, GR.1.4: complementary and supplementary swapped] |
| 3 | Two lines cross. One angle is 47°. Find a and b. | ***a = 47°, b = 133°*** | a = 133°, b = 47° — vertical and adjacent swapped [Math Nation names thinking every adjacent pair has a special sum] |
| 4 | $\angle P$ and $\angle Q$ are complementary, and $m\angle P = 28^\circ$. What is $m\angle Q$ ? | ***B — 62°*** | (A) 180 − 28: supplementary for complementary [Florida B1G-M, GR.1.4: complementary and supplementary swapped]; (C) took the two as equal — that is vertical angles [Notes I]; (D) 90 + 28: added instead of subtracting [Notes II: the two add to 90] |
| 5 | A linear pair measures $(3x + 15)^\circ$ and $105^\circ$. Find x. | ***x = 20*** | 30 — set 3x + 15 = 105, as if vertical [Math Nation names setting a linear pair equal]; −10 — used 90 [Florida B1G-M, GR.1.4] |
| 6 | Vertical angles measure $(6x - 20)^\circ$ and $(2x + 28)^\circ$. Find the measure of each angle. | ***52°*** | 12 — reported x as the angle [Math Nation names reporting x]; 21.5 — added them to 180 [vertical angles are equal, not supplementary] |
| 7 | In the figure, lines AD and BE cross at O, and ray OC is drawn. Which two angles are adjacent? | ***C — angles AOB and BOC*** | (A) a vertical pair: across the crossing, no shared side [Notes I]; (B) angle AOB lies inside angle AOC — the two overlap [Notes I: no overlap]; (D) these share only the vertex O, not a side [Notes I: a shared side] |
| 8 | Above a straight line a ray makes angles of 38° and x°; below it another ray makes an angle of 50°. Find x. | ***x = 142*** | 92 — used the 50° (180 − 38 − 50) [the extra number]; 52 — used 90 [Florida B1G-M, GR.1.4] |
| 9 | Can two vertical angles be complementary? Always, sometimes or never? Explain. | ***Sometimes — only when each is 45°.*** | 'Never — they are not next to each other' [complementary is about the sum, not about being adjacent]; 'always' [Florida B1G-M, GR.1.4] |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** One card on the desk: Corner (90) — complementary; Straight (180) — supplementary; across — vertical, equal; next to, sharing a side — adjacent. Name the pair before writing a number.

**ELL.** **Adjacent**: next to. **Vertical** here means across from (not up and down). **Complement / supplement**: the piece that completes 90 / 180. Point and say each one.

**Enrichment.** Three lines cross at one point and make six angles. Two of them are 40° and 75°. Find the other four. (40°, 75°, 65°, 65°: vertical pairs are equal, and the three around one side make 180.)

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: vertical angles are equal, so they are complementary only when each is 45° — sometimes, never always. A board that says 'yes' or 'no' without the 45 has the verdict and not the reason.

**Vocabulary.** **adjacent angles**; **vertical angles**; **complementary angles**; **supplementary angles**

**Materials.** Whiteboards and markers. Calculators allowed, printed nowhere.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 11 test: Identify complementary, supplementary, vertical, adjacent, and congruent angles — HGV; Find measures of complementary, supplementary, vertical, and adjacent angles — R2B; Write and solve equations using angle relationships — DBV.